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RESEARCH ARTICLE

Enhancing DNN Models for EEG/ECoG BCI With a Novel Data-Driven Offline Optimization Method

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ABSTRACT A better understanding of ElectroEncephaloGraphy (EEG) and ElectroCorticoGram (ECoG) signals would get us closer to comprehending brain functionality, creating new avenues for treating brain abnormalities and developing novel Brain-Computer Interface (BCI)-related applications. Deep Neural Networks (DNN)s have lately been employed with remarkable success to decode EEG/ECoG signals for BCI. However, the optimal architectural/training parameter values in these DNN architectures have yet to receive much attention. In addition, new data-driven optimization methodologies that leverage significant advancements in Machine Learning, such as the Transformer model, have recently been proposed. Because an exhaustive search on all possible architectural/training parameter values of the state-of-the-art DNN model (our baseline model) decoding the motor imagery EEG and finger tension ECoG signals comprising the BCI IV 2a and 4 datasets, respectively, would require prohibitively much time, this paper proposes an offline model-based optimization technique based on the Transformer model for the discovery of the optimal architectural/training parameter values for that model. Our findings indicate that we could pick better values for the baseline model's architectural/training parameters, enhancing the baseline model's performance by up to 14.7% in the BCI IV 2a dataset and by up to 61.0% in the BCI IV 4 dataset.

INDEX TERMS ECoG/EEG signals, BCI, CNN, self-attention model, deep learning, data-driven optimization.

I. INTRODUCTION

ElectroEncephaloGraphy (EEG) records the brain's spontaneous electrical activity as an electrogram. It is ordinarily non-invasive, with electrodes inserted along the scalp (often called "scalp EEG") with the International 10-20 system (see Figure 1) or variants thereof. In addition, EleCtrocorticoGraphy (ECoG) is a form of electrophysiological monitoring that records electrical activity from the cerebral cortex using electrodes placed directly on the brain's exposed surface. Electrodes used in traditional EEG measure this activity from outside the skull. On the other hand, because a craniotomy (a surgical incision into the skull) is necessary to install the electrode grid, ECoG is an invasive procedure (see Figure 2).

The extraction of relevant information from EEG recordings using deep Neural Networks (NN)s has demonstrated promising results in detecting neurological illnesses such as epilepsy [2], [3] and recognizing ischemic stroke [4], identifying sleep disorders [5], decoding motor imagery tasks [6], [7], as well as hand movement preparation stages [8], and supporting stroke rehabilitation via Brain-Computer Interface (BCI) [9]. In addition, the extraction of useful information from ECoG recordings using deep NNs has been proven very effective in imagined 3D continuous hand translation [10], decoding the finger flexion [11], [12] recognizing the state of behavioral sleep and waking state [13], decoding hand gestures [14], [15], and detecting speech activity [16]. All the

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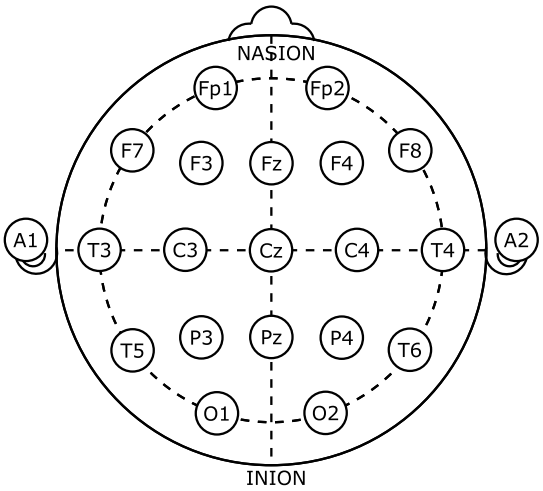

FIGURE 1. Electrode locations of International 10-20 system for EEG (electroencephalography) recording. The "10" and "20" indicate that the distance between neighboring electrodes is either 10% or 20% of the entire front-to-back or right-to-left distance of the skull.

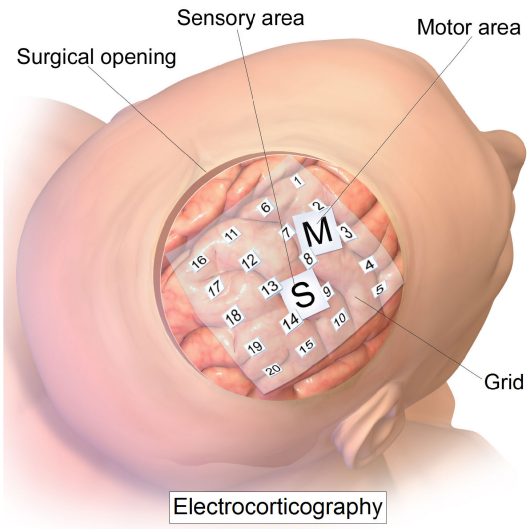

FIGURE 2. Intracranial electrode grid for electrocorticography. See that sensors grid installation requires surgery. Electrodes come out through holes made in the skull and the scalp. Reprinted from [1].

above works extract features from the raw EEG or ECoG signals using DL, most usually Convolutional Neural Networks (CNN)s, and then harness a classification or regression model to make predictions.

However, despite the tremendous advances in deep NNs-assisted EEG/ECoG analysis, optimizing the EEG/ECoG-decoding-NN models' parameters has yet to receive much attention. Because we cannot derive a closed-form formula for the evaluation of a deep NN model (we say that we consider it as a "black box"), and on top of that, the evaluation of a deep NN model is time-consuming because it involves the model training/test on given training/test sets, offline Model-Based Optimization (MBO) is an attractive method for optimizing the EEG-decoding-NN models architectural/training parameters.

In offline MBO, a surrogate model of a complex system[1] (like the evaluation of DNN consisting of many parameters which in turn take many possible values) is first derived and then used as a substitute for the original system (of the DNN model evaluation), usually in an evolutionary algorithm that performs the optimization. As a result, the optimization needs less time, or the evolutionary algorithm can make a more extensive search space exploration, arriving at better solutions.

Offline MBO has proven very effective in biology problems [17], [18], [19]. However, all previous works mitigate the wrong guidance given by the surrogate model to the optimizer due to inaccurate predictions, more often overestimations. That happens because the estimated model is trained on a small manifold of the possible input space, making the predictions far from the training distribution harder. More specifically, [17] leverages a maximum-likelihood (NML) estimator to reason about uncertainty and out-of-distribution values and guide the optimizer to the best solution with more confidence. In addition, [18] proposes a model-based adaptive sampling, to sample with larger probability inputs that may better optimize the desired properties during optimization. Finally, [19] introduces the conservative objective models (COM)s, i.e., a method that learns a model of the objective function that lower bounds the actual value of the ground-truth objective on out-of-distribution inputs and uses it for optimization.

In this paper, we exploit a data-driven offline optimization technique based on the Transformer model to optimize the state-of-the-art deep CNN model decoding the motor imagery EEG signals in the BCI IV 2a datasets [20] and the finger tension ECoG signals in the BCI IV 4 dataset [21]. In order to do that, we first perform a random sampling on the baseline model architectural/training parameters space under consideration to create a dataset that matches these random architectural/training parameter values to the evaluated accuracy of the baseline model. Then, we provide that dataset to the proposed model-based optimization methodology that first builds a surrogate model of the baseline model accuracy given architectural/training parameter values and then makes use of that surrogate model in a metaheuristic optimization algorithm to find the architectural/training parameter values that achieve the best accuracy. Our main contributions in this work are:

1) Improve the accuracy of an existing Deep Learning (DL) model for decoding EEG & ECoG signals of BCI applications.
2) Enhance the generalizability of the surrogate model in the proposed MBO method to optimize the DDN models for decoding EEG/ECoG signals of new unseen subjects.

[1] A system that requires a long time to respond at a given input and the input space is gigantic.

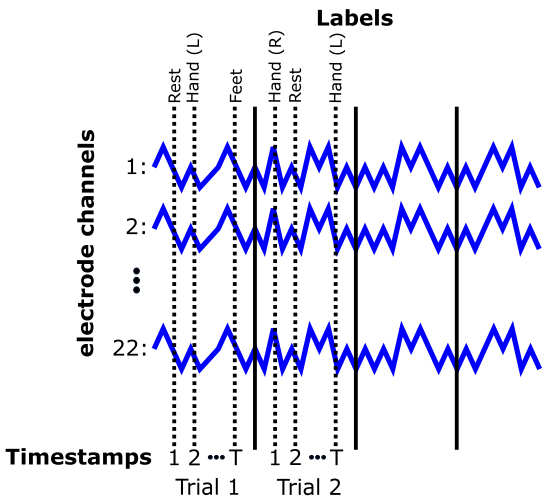

FIGURE 3. A pictorial representation of the BCI IV 2a dataset. The time is divided into trials, and each trial is divided into timestamps. One of the labels Rest, Feet, Hand (L), and Hand (R) is given to every timestamp.

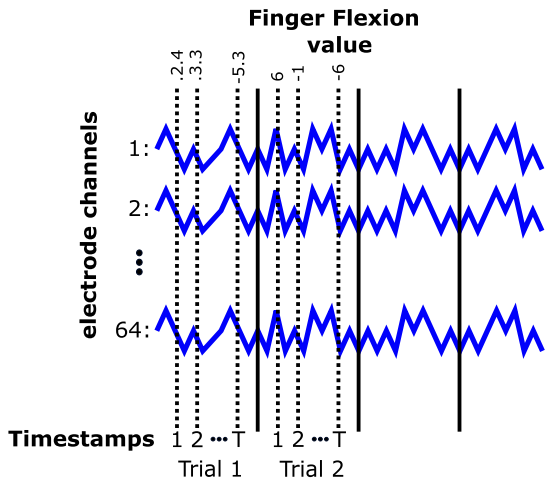

FIGURE 4. A pictorial representation of the BCI IV 4 dataset. The time is divided into trials, and each trial is divided into timestamps. A finger flexion value, i.e., a real continuous value, is given to every timestamp.

3) Show that the proposed data-driven MBO technique is better than other MBO techniques for the optimization of DNN models decoding EEG/ECoG signals.
4) Present an optimization paradigm for also optimizing the DNN models decoding other bio-signals.

The rest of the paper is organized as follows: Section II describes the baseline deep CNN model architecture and the motor imagery EEG and finger tension ECoG signals datasets. Section III explains the proposed data-driven offline optimization methodology to find the best values for the baseline CNN model architectural/training parameters. Finally, in section IV, we demonstrate the better accuracy we can obtain using the architectural/training values found by the proposed optimization methodology while conclusions are drawn, and our plans for future work are presented in Section V.

II. EEG/ECoG SIGNALS DECODING

The authors of [22] demonstrated a variety of CNN architectures, ranging from 2- to 31-layer CNN networks, for decoding the EEG signals in the motor imagery BCI IV 2a dataset. Furthermore, these deep CNN models could decode motor imagery EEG signals as well as or better than Filter Bank Common Spatial Patterns (FBCSP) [23], the winner of many EEG-decoding competitions. Finally, the Shallow ConvNet model was the most accurate on average for all subjects among all the proposed architectures in that work.

A. BCI IV 2a DATASET

The 22-electrode EEG recordings comprising the BCI IV 2a dataset represent nine patients' motor imagery-related brain activity. Each individual i has a distinct collection of EEG recordings made while performing four distinct body movements, namely *"Left Hand"*, *"Right Hand"*, *"Feet"* and *"Rest"*. The j-th entry of the i-th subject in the dataset is represented as $D^i = \{X^j, Y^j\}$, j from 1 to the total number of recorded trials, where $X^j \in \Re^{E\times T}$ and $Y^j \in \Re^T$, with E denoting the number of recorded electrodes, T the number of timesteps after the discretization of each trial, and Y^j being labeled with one of the four types of body movements, in each timestep of the j-th trial (see Figure 3).

B. BCI IV 4 DATASET

The shallow ConvNet model has also been tested on Electrophysiological (ECoG) signals in the BCI IV 4 dataset showing better performance than previous works.

The BCI IV 4 dataset consists of 62, 48, and 64 -electrode ECoG recordings corresponding to 3 epileptic patients. The dataset has the same structure as the BCI IV 2a. However, measurements of finger flexion (continuous values) are given at every timestep instead of class labels (see Figure 4).

C. SHALLOW ConvNet

Shallow ConvNet mimics the spatial filters (bandpass and Common Spatial Patterns (CSP)) in FBCSP with temporal and spatial convolution in the initial two layers. A squaring nonlinearity layer, a mean pooling layer, and a dense layer with the softmax activation function comprise the remainder of the pipeline, which resembles the logarithmic variance calculation in FBCSP. Figure 5 illustrates the Shallow ConvNet model architecture. Although Shallow ConvNet is accurate enough, its accuracy can be improved further by tuning its architectural/training parameters.

III. DATA-DRIVEN OFFLINE OPTIMIZATION OF THE SHALLOW ConvNet ARCHITECTURE

To improve the Shallow ConvNet architecture, we alter some of the Shallow ConvNet architectural/training parameters. First, we assume a vast search space consisting of 8.5 billion

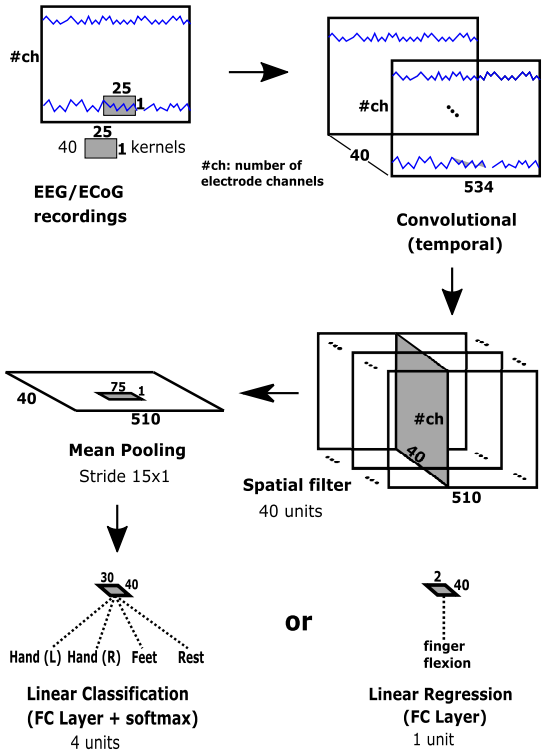

FIGURE 5. The Shallow ConvNet model architecture decoding the BCI IV 2a or 4 datasets, respectively. The model comprises a temporal convolutional layer, i.e., convolutional operations through time with 40 different kernels, a spatial filter, i.e., convolutional operations across the 40 channels for every timestamp, a mean pooling layer, and finally, a linear classification layer with 4 outputs when the model is decoding the BCI IV 2a dataset (EEG) or a linear regression layer when the model is decoding the BCI IV 4 dataset (ECoG).

distinct architectural/training configurations to exhaust our options for finding a more accurate Shallow ConvNet-like model. Then, using a small sample of Shallow ConvNet-like modified models, $\mathcal{D} = \{\{x_1, y_1\}, .., \{x_i, y_i\}, \ldots, \{x_n, y_n\}\}$, with x_i being a Shallow ConvNet architecture with random architectural/training parameters values, and y_i being the average accuracy across all subjects of that model after training and testing it on a given training/test set, we train a surrogate model, based on the Transformer model, predicting the accuracy of new Shallow ConvNet architectures. Finally, we employ the Firefly evolutionary optimization algorithm using the previous surrogate model to explore the Shallow ConvNet architectural/training space and discover their best values that maximize the accuracy of the Shallow ConvNet model [22].

A. SHALLOW ConvNet ARCHITECTURAL/TRAINING PARAMETERS SEARCH SPACE

This subsection discusses the architectural/training Shallow ConvNet design choices we consider in our study. A Shallow ConvNet-like architectural model and its training are defined by eight discrete-valued variables (p_i for i from 1 to 8), each corresponding to a different Shallow ConvNet architectural/training parameter. As shown in Table 1, the first five

TABLE 1. Architectural/training parameters of shallow convnet defining our search space.

Architectural/Training parameter p_i	Num. of discrete values	Actual range of values *[Start:End:Step]*
Temporal Filters	30	[25 : 54 : 1]
Temporal Filter Length	26	[10 : 35 : 1]
Spatial Filters	29	[25 : 53 : 1]
Temporal Pool Kernel Length	33	[50 : 82 : 1]
Temporal Pool Stride Length	20	[10 : 29 : 1]
Dropout Probability	7	[0.2 : 0.8 : 0.1]
Learning Rate	9	[1.25e-2 : 9.25e-2 : 1e-2]
Weight Decay	9	[1.125e-4 : 9e-4 : 1.125e-4]

design choices refer to the Shallow ConvNet architectural model parameters (see Fig. 5), and the last three are associated with tuning the training of the Shallow ConvNet model.

B. SURROGATE MODEL

The purpose of the surrogate model is to predict the accuracy of the Shallow ConvNet model for any combination of the architectural/training parameters shown in Table 1. The objective in the training of the surrogate model is essentially the minimization of the mean-squared-error between the projected $f_\theta(x_i)$ and the ground truth y_i accuracy values. However, such a model can overestimate its predictions at unseen configurations (call them x^- configurations), as shown in [19]. Therefore, we optimize f_θ to discover those unseen configurations (x^-) and formulate the objective function appropriately to prevent f_θ from learning overestimated values at x^- configurations [24]. Thus, the objective in training is:

$$\theta^\star = \underset{\theta}{argmin}\,(\mathbb{E}_{xi,yi\sim D}[(f_\theta(x_i) - y_i)^2] - \alpha\,\mathbb{E}_{x^-\sim Opt(f_\theta)}[f_\theta(x^-)]), \quad (1)$$

where $\mathbb{E}_{xi,yi\sim D}[(f_\theta(x_i) - y_i)^2]$ is the MSE term, $\alpha\,\mathbb{E}_{x^-\sim Opt(f_\theta)}[f_\theta(x^-)]$ is the term in the objective to prevent overestimations on unseen configurations, and $Opt(f_\theta)$ is the Optimizer, e.g., the Gradient Ascent (GA), to discover the x^- configurations by maximizing the surrogate model (f_θ).

1) CONTEXT-AWARE SURROGATE MODEL

In order to make the surrogate model more general and can accommodate new subjects without requiring to re-train it, we extended its modeling with a context vector $c_k \in \Re^5$ that contains information (features) about the subjects. In our study, we consider the age group and the gender of the subjects [2] as features. More specifically, the subjects' age group can be 18 to 30, 31 to 50, or 50+, and their gender can be male or female. Because that information is unavailable in the datasets under test, we assume random values for these features. Therefore, we add semantic information to the surrogate model, which is harnessed by the optimization algorithm, we describe in subsection III-D, to discover either the

[2] One can also think of other helpful information to add, e.g., information related to the subject's medical history.

best-accuracy on average ConvNet-like architecture across all subjects, i.e., what we do in case we do not consider any context vector with the subtle difference that in the context-aware case the x_is in the dataset are annotated with the Shallow ConvNet's accuracy of the subject under test each time and not the average accuracy across all subjects because the context vector offers the degree of freedom to do so or the best-accuracy ConvNet-like architecture for a novel context vector pertinent to a new subject (personalized healthcare). In the latter case, the surrogate model is not re-trained on any data corresponding to the new subject, which is expensive, but the relevant information about the new subject is conveyed into the surrogate model through the context vector. The optimization problem for the training of the surrogate model is now defined as follows:

$$\theta^{\star} = \underset{\theta}{argmin}\,(\mathbb{E}_{\substack{x_i, y_{i,k} \sim \mathcal{D}_n \\ k \sim [K]}}[(f_\theta(x_i, c_k) - y_{i,k})^2] - \alpha \mathbb{E}_{\substack{x^- \sim Opt(f_\theta) \\ k \sim [K]}}[f_\theta(x^-, c_k)]). \quad (2)$$

Specifically, we built a dataset $\mathcal{D} = \{\mathcal{D}_1 \ldots, \mathcal{D}_N\}$, where each $\mathcal{D}_i$ refers to a ConvNet-like architecture configured with randomly selected values of the architectural/training parameters in Table 1 (x_i) and annotated with its decoding accuracy on the test-set of the k-th subject's recordings ($y_{i,k}$). Also, c_k is the context vector for the k-th subject. The main difference between eq. (1) and eq. (2) is that in eq. (1) y_i is the average Shallow ConvNet's accuracy across all subjects given x_i, while in eq. (2) y_{i,c_k} is the Shallow ConvNet's accuracy of the k-th subject given x_i, and f_θ depends besides x_i to c_k too.

The context-aware DL pipeline that describes the surrogate model ($f_\theta(x_i, c_k)$) is shown in Figure 6. The one-hot encodings of the architectural/training parameters are stacked together into a 163-dimensional vector ,[3] which is then embedded into a 512-dimensional vector space. The previous embedding is then unpacked into 8×64 vectors corresponding to each architectural/training parameter. After that, a "positional encoding" of the parameters is added to the input embedding. The positional encodings have the same dimension as the input embeddings, i.e., 64 dimensions, so the two can be summed. We consider the sine-and-cosine-with-different-frequencies-positional-encoding [25], shown below:

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/64}),$$
$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/64}) \quad (3)$$

where pos is the parameter's position in the list of parameters and takes values from 0 to 7, and i is the dimension in the embedding, and takes values from 0 to 31. The previous embeddings then go through 2 Multi-Head Attention [25] layers before the output of 8×64 is flattened into a vector of size (512,), i.e., the Transformer embeddings. In parallel, the one-hot encodings of the context vectors are stacked together in a 5-dimensional[4] vector which is then embedded into two different 512-dimensional vector spaces, one through one dense layer and the other through two dense layers. After that, the first context vector embedding, i.e., the embedding through one dense layer, is cross-multiplied with the Transformer embeddings. Then, the 512-dim point-wise multiplication product is fed into N different voting prediction sub-networks, spawning $f_\theta^1(x, c_k), f_\theta^2(x, c_k), \ldots, f_\theta^N(x, c_k)$. In addition, the 512-dim point-wise multiplication product is concatenated with the second 512-dim context vector embedding, i.e., the context vector embedding through the two dense layers, to create a 1024-dim vector, which is given as input to two dense layers producing N weights $w_1, w_2, \ldots, w_N$ that decide the contribution of each f_θ^i. The final output is obtained by $f_\theta(x, c_k) = \sum_{i=1}^{N} w_i f_\theta^i(x, c_k)$, where $\sum_{i=1}^{N} w_i = 1$.

Finally, the training of the context-aware surrogate model is performed as described in Algorithm 1.

Algorithm 1 Surrogate Model Training

```
Initialize the DL model of f_θ(x, c_k)
for every batch in D do
    Initialize x^- values with the x values in the batch
    for _ in range: Opt_iter do
        Update the x_j^- values to maximize f_θ(x), ∀j
    end for
    Update θ using the Adam Optimizer with Loss
    function the Eq. 2 and negative samples the x^- values
end for
return final learned model f_θ
```

C. ATTENTION MECHANISM

Our surrogate model leverages the Multi-Head self-Attention (or intra-attention) mechanism proposed in [25]. Intuitively, the self-Attention mechanism assesses the semantic relationship between every Shallow ConvNet architectural/training parameter with every other architectural/training parameter in Table 1, including itself, by assigning higher weights when the semantic relationship is stronger and lower weights when the semantic relationship is weaker and then, re-scaling the initial parameters' embeddings according to these weights. Therefore, the result is eight embedding vectors that carry contextual information. For example, the "Temporal Filters" parameter in Table 1 would be compared with itself and the rest (other seven) of the architectural/training parameters to compute the weights showing its semantic relationship with the other parameters, and its embedding would be updated by scaling the embeddings of the other parameters according to these weights.

[3]The sum of the discrete values the architectural/training parameters can obtain.

[4]Three dimensions corresponding to the subjects' age group, and 2 dimensions corresponding to the subjects' sex.

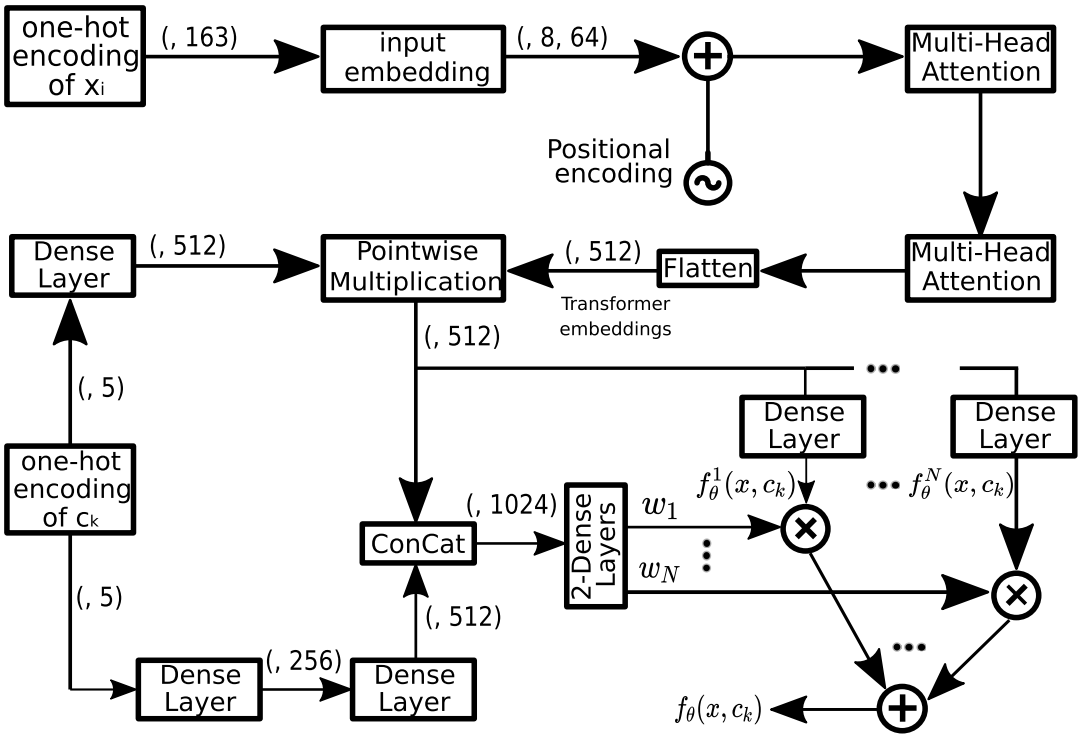

FIGURE 6. The DL pipeline of the surrogate model. Two Multi-Head Attention layers embed the architectural/training parameters into a vector space. Then, these embeddings are fused with embeddings of context vectors pertinent to each subject. The resulting embeddings from the fusion are 1) used to find suitable functions $f_\theta^1, \ldots, f_\theta^N$, and 2) weights $w_1, \ldots, w_N$ for scaling $f_\theta^1, \ldots, f_\theta^N$, respectively, so that after summing all the products of the functions $f_\theta^1, \ldots, f_\theta^N$ with the respective weights to arrive at the final surrogate model.

1) SCALED DOT-PRODUCT ATTENTION

In more detail, assume the eight initial Shallow ConvNet architectural/training parameters embeddings be $\{\mathbf{u}_i\}_{i=1\ldots8} \in \Re^{1\times d_k}$, and define the "Query" matrix $\mathbf{Q} \in \Re^{d_k\times d_k}$, the "Key" matrix $\mathbf{K} \in \Re^{d_k\times d_k}$, and the "Value" matrix $\mathbf{V} \in \Re^{d_k\times d_k}$ with the corresponding i-th row of these matrices being

$$q_i = \mathbf{u}_i\mathbf{M}_q + \mathbf{b}_q$$
$$k_i = \mathbf{u}_i\mathbf{M}_k + \mathbf{b}_k$$
$$v_i = \mathbf{u}_i\mathbf{M}_v + \mathbf{b}_v,$$

respectively, where the $\mathbf{M}_q$, $\mathbf{M}_k$, $\mathbf{M}_v$ matrices $\in \Re^{d_k\times d_k}$, and the $\mathbf{b}_q$, $\mathbf{b}_v$, and $\mathbf{b}_v$ bias vectors $\in \Re^{1\times d_k}$, are going to be learned during the self-Attention model training. Then, the final parameters embeddings are derived by performing a weighted-sum over all the initial embeddings ($\mathbf{u}_i$): $\tilde{\mathbf{u}}_i = \sum_{j=1}^{8} w_{i,j}\mathbf{u}_j$ with $w_{i,j}$ being the (i, j) element of the matrix $\mathbf{W}$ defined as:

$$\mathbf{W} = softmax(\frac{\mathbf{QK}^T}{\sqrt{d_k}}).$$

The aforementioned attention mechanism is called Scaled Dot-Product Attention in [25]. See that the multiplication $\mathbf{QK}^T$ essentially measures the semantic relationship or the similarity between all the parameters i.e. the dot product between every row of the matrix $\mathbf{Q}$ (I remind you that every row of the matrix $\mathbf{Q}$ correspond to an architectural/training parameter) with the columns of the matrix $\mathbf{K}^T$ (again every column of the matrix $\mathbf{K}^T$ corresponds to an architectural/training parameter). Finally, the element-wise multiplication with the $\frac{1}{\sqrt{d_k}}$ mitigates the vanishing gradient issue that could make the model training unstable.

2) MULTI-HEAD ATTENTION

Now, instead of having a single self-attention layer, we can split this operation into h parallel self-attention layers/heads, each generating $\frac{d_k}{h}$ dimensional contextual representations. The "Query", "Key" and "Value" matrices are first given as input to h parallel Linear layers. Then, the outputs of the self-Attention heads are concatenated altogether before being passed to a new Linear Layer. The multi-head mechanism enables the attention model to jointly attend to different semantic information between the architectural/training parameters. Doing so allows the discovery of more powerful contextual embeddings, compared to the single-head paradigm.

D. FIREFLY OPTIMIZER

After training the surrogate model with Algorithm 1, we call the Firefly optimizer [26], [27], [28] to start searching for the architectural/training values that lead to the most accurate Shallow ConvNet architecture. Initially, a population of

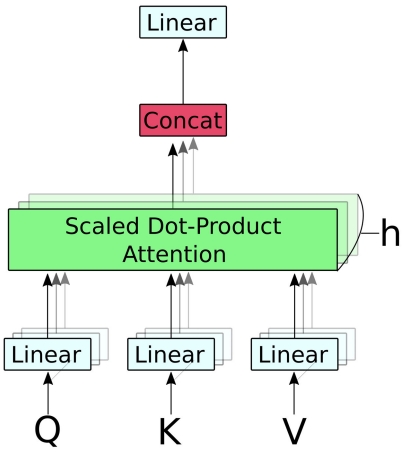

FIGURE 7. Multi-Head Attention model. See the h parallel Scaled Dot-Product Attention layers, each finding a $\frac{d_k}{h}$-dimensional embedding, and d_k being the dimension of the final embedding.

n different firefly particles, i.e., the Shallow ConvNet Architecture with different parameter values, is selected. Each firefly particle x_i in the population is associated with an intensity value I, i.e., $f_\theta(x_i)$. The intuition behind updating the "position" of a firefly is based on the observation that a firefly with higher intensity I_j located at position x_j will attract another firefly with a smaller intensity I_i located at position x_i. The position update step is performed as follows:

$$x_i^{t+1} = x_i^t + e^{-r_{ij}^2}(x_j^t - x_i^t) + \alpha_t \epsilon_i^t, \tag{4}$$

with

$$\alpha_t = 0.97\alpha_{t-1},$$

where r_{ij} is the intensity distance between the firefly particle i and the firefly particle j, ϵ_i^t is a random configuration from the design space, and α_t is a changing parameter in every iteration that controls the diversity of the design solution. Moreover, because all the calculations in Equation 4 involve continuous variables, but the Shallow ConvNet architectural/training parameters we consider are discrete variables, we need to map the continuous values to discrete values. Therefore, after performing the calculations in the position update step, we map the updated position x_i^{t+1} to the closest valid discrete value. Then, after repeating the update step for C cycles, the final list of firefly particles is returned as the solution to the optimization problem, which in our case is the Shallow ConvNet architectural/training parameter values that lead to the best accuracy (see Algorithm 2).

IV. EXPERIMENTAL RESULTS

In this section, we demonstrate that the Firefly Optimizer, using as objective the proposed surrogate model in subsection III-B, could discover the best values of the architectural/training parameters in Table 1 that lead to a more accurate Shallow ConvNet model, tested on the BCI IV 2a and 4 datasets.

Algorithm 2 Firefly Pseudocode

Input: Surrogate model $f_\theta(x)$
Initialize the firefly population x_i, $i = 1, 2, \cdots, n$, i.e., random combinations of the Shallow ConvNet model architectural/training values.
$I_i = f_\theta(x_i)$, $i = 1, 2, \cdots, n$
for $t = 1 : C$ **do**
  for $i = 1 : n$ **do**
    for $j = 1 : n$ **do**
      if $I_i^t < I_j^t$ **then**
        Update x_i^t (to x_i^{t+1}) with Eq. 4
        $I_i^{t+1} = f_\theta(x_i^{t+1})$
        if $I_i^{t+1} < I_i^t$ **then**
          $x_i^{t+1} = x_i^t$
          $I_i^{t+1} = I_i^t$
        end if
      end if
    end for
  end for
end for
return $\underset{x_i, \forall i}{argmax}\, f_\theta(x_i)$

Our experimental framework was developed in Python using the TensorFlow library. All our experiments were performed on a system with a single NVIDIA V100 with 64GB of RAM, and a 3.7GHz Intel Xeon Gold 6154 CPU.

A. TRAINING CONFIGURATIONS

1) CONTEXT-LESS APPROACH

Initially, we created three offline datasets, one for the BCI IV ECoG regression challenge and the other for the BCI IV 2a EEG high and low-frequency classification challenges ,[5] respectively, to train the surrogate models of our proposed method. Each dataset consists of 8000 samples, and each sample results from a different random combination of the Shallow ConvNet architectural/training parameter values shown in Table 1. In addition, every sample in the offline dataset is annotated with its average decoding accuracy across all subjects, obtained by re-training the Shallow ConvNet using the random architecture/training parameters values associated with that sample itself.

For each offline dataset $\mathcal{D}$, created in the previous step, we have trained a surrogate model using Algorithm 1 for approximately 530 epochs. We considered the following hold-out validation strategies for each test case during our experiments. Either train on 80% of the configurations in the offline dataset and validate on the rest 20%, or train on the 90% of the configurations and validate on the rest 10%. This approach leads to finding a Shallow ConvNet model's ideal architectural and training settings with optimal

[5]The EEG signals in the BCI IV 2a dataset are filtered with a 3rd order Butterworth filter, resulting in a low (0-38Hz) and a high (4-38Hz) frequency sub-datasets.

decoding accuracy across all subjects in the examined test case.

2) CONTEXT-AWARE APPROACH

Similarly, we randomly permuted the available architectural/training parameter values in Table 1 to create three offline datasets with 8000 different Shallow ConvNet architectures. This time, each sample in the offline datasets is annotated with the same number of individual decoding accuracy metrics as the number of available subjects recordings under examination, i.e., nine and three in the BCI IV 2a and BCI IV 4 test cases, respectively. As in IV-A1, obtaining the decoding accuracy metric associated with a sample in one of the offline datasets requires the Shallow ConvNet to be re-trained using the random architecture/training parameters values associated with that sample.

Each offline dataset is then used to train a context-aware surrogate model by minimizing Eq. 2 with Algorithm 1 for 1060 and 1590 epochs in ECoG-signals-decoding-model and EEG-signals-decoding-model test cases, respectively. We have considered two strategies during our evaluation; One that optimizes for unseen subjects by leaving one subject out from the training data (including validation split) and the other one that allows the surrogate to be trained on all available data we have created corresponding to all subjects (refer to III-B1). We employed the same hold-out validation strategies for these two evaluation methods as in the context-less approach (IV-A1).

B. DISCOVERED MODELS VS ConvNet MODEL

Tables 2 and 3 compare the baseline ConvNet model to the highest-performing models derived from the context-less approach (described in IV-A1), on both high and low-frequency test cases in BCI IV 2a EEG challenge, and on the single-frequency test case in BCI IV 4 ECoG challenge, respectively. Table 4 depicts the optimum architectural/training parameter values, which attain the decoding performance scores reported in Tables 2 and 3, discovered by the Firefly Optimizer (refer to III-D) using the context-less f_θ as objective. It is clear that the data-driven optimization, leveraging offline data, can find a better combination of the ConvNet architectural/training parameter values than the default values, which leads to better metrics in decoding the EEG/ECoG signals recordings in the BCI challenge datasets.

Tables 5 and 6 illustrate the highest-performing models obtained when we optimize for a single novel context vector relevant to an unseen subject. The values in the tables present the decoding accuracies of the "excluded" subject from the offline training data. It is apparent that the context-aware modeling of the surrogate model allows our optimizer (refer III-D) to discover architectures that outperform the baseline ConvNet by a significant margin. It is worth mentioning that besides improving the individual decoding metrics associated with the discovered architectures of the left-out subject, the context-aware approach also obtains state-of-art average

TABLE 2. Top decoding-test average accuracies, across all patients, of discovered models VS baseline shallow convnet model on BCI-IV 2a challenge – context-less optimization.

Freq.	Our Model	Shallow ConvNet	D **Best in Training**	D **Mean in Training**
Low	77.2%	73.7%	71.7%	69.2%
High	75.7%	71.9%	70.2%	69.1%
Mean Improvement	**×1.0**	**↑3.4%**	**↑5.5%**	**↑7.3%**

decoding performance. Specifically, we obtained 0.886 average decoding correlation on the ECoG dataset and 75.1% and 75.2% average decoding accuracy on the EEG low and high-frequency test cases, respectively. Note that, during our experiments, we excluded different subjects to make sure our approach was not biased. Our model performance is consistently optimal, independently of the "excluded" subject selection. In detail, the individual decoding accuracies reported in Table 5 refer to the excluded subjects with context IDs 2 and 4 for the low and high frequencies EEG test cases, respectively. Similarly, in Table 6, we have excluded the subject with context ID 2. In Table 7, we annotate the ideal architectural/training parameter values for the Shallow ConvNet-like architectures that achieve the best decoding accuracy on each BCI challenge, obtained by optimizing for a novel context vector relevant to a new subject.

Tables 8 and 9 demonstrate the best-performing models obtained when we optimize simultaneously across all available subjects without keeping out any subject recordings from our offline training data. We call this strategy multi-context optimization. Adopting this optimization strategy led to the discovery of ConvNet-like architectures, which exhibit the best decoding metrics during our study on all BCI IV datasets, considered in this study. Table 10 contains the optimal architectural/training parameter values for attaining the decoding performance metrics reported in Tables 8 and 9.

It is essential to mention that the reported accuracy and correlation scores in all tables in this section refer to the ground truth accuracy/correlation achieved by re-training the ConvNet model on the respective BCI IV dataset, using the optimal architecture and training parameters[6] discovered by each data-driven optimization variant, presented in this work.

In addition, the proposed optimizations ran $\times 7.87 * 10^6$ and $\times 4.7 * 10^6$ days faster, on average, than an exhaustive search of all possible architectural/training parameter values for the BCI IV 2a high-frequency and low-frequency test cases, respectively. Similarly, for the BCI IV 4 ECoG test case, our optimizations achieved an average speedup of $\times 9.8 * 10^6$ days, which means we can tune the model on different datasets without relying on the default architectural/training parameter values. Finally, it is worth mentioning that our

[6]Optimal under our learned f_θ, trained on the offline datasets (see IV-A).

TABLE 3. Top decoding-test average accuracies, across all patients, of discovered models VS baseline shallow convnet model on BCI-IV 4 ECoG challenge – context-less optimization.

Freq.	Our Model	Shallow ConvNet	*D* **Best in Training**	*D* **Mean in Training**
1-200 Hz	0.87	0.67	0.73	0.43
Improvement	**×1.0**	**↑29.8%**	**↑19.1%**	**↑102.3%**

TABLE 4. Architectural and training parameters of the top performing models discovered for each BCI test case – context-less optimization.

Arch./Training Params.	BCI IV 2a Low Freq.	BCI IV 2a High Freq.	BCI IV 4
Temporal Filters	25	36	27
Temporal Filter Length	31	30	35
Spatial Filters	49	31	48
Temporal Pool Kernel Filter	62	51	82
Temporal Pool Stride Length	28	29	20
Dropout Prob.	0.8	0.8	0.2
Learning Rate	0.0125	0.0225	0.0125
Weight Decay	23e-5	0.0	68e-5

TABLE 5. Top decoding-test accuracies, associated with the excluded subject from the training data, of discovered models VS baseline shallow ConvNet model on BCI-IV 2a challenge – unseen subject optimization.

Excluded Subj. ID	Freq.	Our Model	Shallow ConvNet	*D* **Best in Training**	*D* **Mean in Training**
2	Low	59.4%	49.3%	47.1%	40.7%
4	High	75.0%	68.8%	66.9%	61.2%
	Mean Improvement	**×1.0**	**↑14.7%**	**↑19.1%**	**↑34.2%**

TABLE 6. Top Decoding-Test correlation scores, associated with the excluded subject from the training data, of discovered models VS baseline Shallow ConvNet model on BCI-IV 4 ECoG challenge – unseen subject optimization.

Excluded Subj. ID	Freq.	Our Model	Shallow ConvNet	*D* **Best in Training**	*D* **Mean in Training**
2	1-200 Hz	0.95	0.59	0.52	0.49
	Mean Improvement	**×1.0**	**↑61.0%**	**↑82.6%**	**↑93.8%**

approach identified more accurate Shallow ConvNet architectures despite training on design choices associated with decoding performance below the baseline (see the columns "*D* Best/Mean in Training"[7] of Tables 2, 5, 6, and 8).

C. COMPARISONS WITH OTHER DATA-DRIVEN OFFLINE MBO METHODS

The authors of Design-Bench [29] introduced a benchmark for Data-Driven Offline Model-based Optimizations on a

[7] "*D* Best/Mean in Training" columns of the previously mentioned Tables correspond to the best/mean accuracy obtained on the generated offline datasets in each case.

TABLE 7. Architectural and training parameters of the top performing models discovered for each BCI test case – unseen subject optimization.

Arch./Training Params.	BCI IV 2a Low Freq.	BCI IV 2a High Freq.	BCI IV 4
Temporal Filters	34	49	31
Temporal Filter Length	13	30	34
Spatial Filters	46	34	53
Temporal Pool Kernel Filter	54	57	81
Temporal Pool Stride Length	28	29	28
Dropout Prob.	0.7	0.8	0.2
Learning Rate	0.0225	0.0125	0.0225
Weight Decay	67.5e-5	67.5e-5	79e-5

TABLE 8. Top Decoding-Test accuracies of discovered models VS baseline shallow ConvNet model on BCI-IV 2a challenge – multi-context optimization.

Freq.	Our Model	Shallow ConvNet	*D* **Best in Training**	*D* **Mean in Training**
Low	77.8%	73.7%	71.7%	69.2%
High	75.2%	71.9%	70.2%	69.1%
Mean Improvement	**×1.0**	**↑5.0%**	**↑7.8%**	**↑10.6%**

TABLE 9. Top decoding-test correlation scores of discovered models VS baseline shallow ConvNet model on BCI-IV 4 ECoG challenge – multi-context optimization.

Freq.	Our Model	Shallow ConvNet	*D* **Best in Training**	*D* **Mean in Training**
1-200 Hz	0.88	0.67	0.73	0.43
Improvement	**×1.0**	**↑31.3%**	**↑20.5%**	**↑104.6%**

TABLE 10. Architectural and training parameters of the top performing models discovered for each BCI test case – multi-context optimization.

Arch./Training Params.	BCI IV 2a Low Freq.	BCI IV 2a High Freq.	BCI IV 4
Temporal Filters	34	49	25
Temporal Filter Length	35	30	10
Spatial Filters	50	34	52
Temporal Pool Kernel Filter	67	57	81
Temporal Pool Stride Length	29	29	10
Dropout Prob.	0.8	0.8	0.2
Learning Rate	0.0125	0.0125	0.0125
Weight Decay	56.25e-5	67.5e-5	79e-5

list of practical tasks derived from real-world challenges, such as designing proteins robots and DNA sequencing. Furthermore, they have publicly disclosed implementing these State-of-the-Art MBO methods, which we utilize in this study as benchmarks for our proposed Data-Driven MBO approach.

Table 11 compares the best-performing architectures discovered by each MBO method, tested on the BCI IV 2a EEG and BCI IV 4 ECoG challenges. The results indicate that our approach prevails in designing ConvNets with higher decoding accuracy metrics in all BCI test cases. Furthermore, the

TABLE 11. Top decoding-test scores derived by the considered offline data-driven MBOs on each BCI test case. element values in bold indicate the best performing method for the respective BCI challenge dataset.

Dataset	Our Method	COMs [19]	CbAS [18]	BO-qEI [30], [31]
ECoG	**0.88**	0.84	0.85	0.81
EEG-Low	**77.8%**	75.6%	73.85%	72.7%
EEG-High	**75.7%**	73.59%	72.3%	70.9%

considered methods used for comparisons [18], [19], [30], [31] have their surrogate models trained on the same offline dataset as the one created for the training needs of our surrogate models, as described in IV-A.

More specifically, the advantage of our method and [19] over [18], [30], [31], is that they mitigate the problem of being over-optimistic at data points that lie outside the distribution of the data samples for building the surrogate model. In addition, the proposed model performs better than [19] because it relies on the Multi-Head attention model that can extract the semantic relationship between the parameters and arrive at better vector embeddings.

D. STATISTICAL VALIDATION

The surrogate model is built on a statistical sample. Because of that, we expect that the predictions for data points that lie far away from the data samples used for building the surrogate model are not going to be too accurate, even though the proposed surrogate model is conservative and mitigates to some extent the problem of inaccurate predictions for those data points. Thus, we are more interested in preserving the slope of the actual function between two data points in the surrogate model, which means if $f(x_1) < f(x_2)$, then we would also like $f_\theta(x_1) < f_\theta(x_2)$ because we are looking for the best parameter values that optimize f. By doing so, instead of optimizing the actual f, which is expensive to evaluate, we can evaluate f_θ. Because of this, instead of providing the MSE between the true f and the predicted f (f_θ) that demands the difference between the two to be as minor as possible, we provide in our experiments the Kendall ranking coefficient that measures how effectively the slope of f between two data point is preserved in f_θ. Consider the set $S = \{y_1, y_2, \ldots, y_i, \ldots, y_n\}$ which represents the ground truth decoding accuracy values on the validation set, and the set $S' = \{y'_1, y'_2, \ldots, y'_i, \ldots, y_n\}$, which represents the predicted decoding accuracy values on the validation set. Then, the Kendall ranking coefficient (τ) is defined as follows:

$$\tau = \frac{c - d}{\binom{n}{2}},$$

where c is the number of pairs in S' that preserve the slope, i.e., the sign of the difference between the two pair elements, of the corresponding pairs in S and is given by

$$c = \sum_i \sum_{i<j} \mathbb{1}[\mathrm{sgn}(y_i - y_j) = \mathrm{sgn}(y'_i - y'_j)],$$

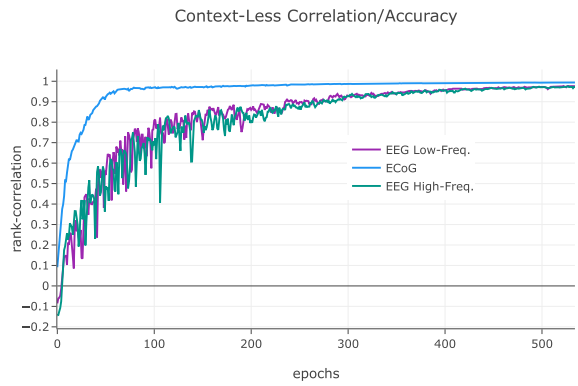

FIGURE 8. Training Curves of our Context-Less surrogate model for each BCI test case. The annotated performance in the training curves were achieved using the following α values: 0.1 on EcoG test case and 0.01 on both EEG test cases. After approximately 530 epochs, the performance index plateaus, at that point the surrogate model has learned to extract meaningful patterns from the offline dataset of each BCI test case.

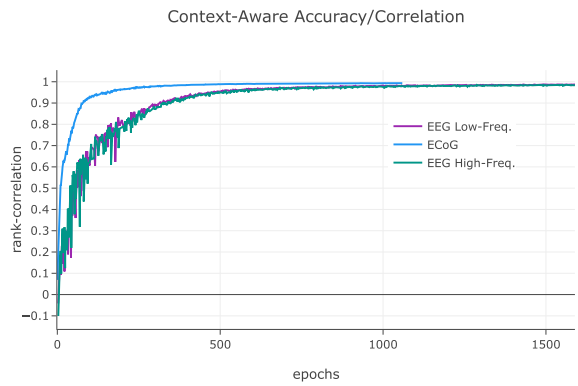

FIGURE 9. Training Curves of our Context-Aware surrogate model for BCI test cases. The annotated performance in the training curves were achieved using the following α values: 0.5 on EcoG test case and 0.1 on both EEG test cases. After approximately 1060 epochs, the performance index plateaus, at that point the surrogate model has learned to extract meaningful patterns from the offline dataset of each BCI test case.

d is the number of pairs in S' that do not preserve the slope of the corresponding pairs in S and is given by

$$d = \sum_i \sum_{i<j} \mathbb{1}[\mathrm{sgn}(y_i - y_j) = -\mathrm{sgn}(y'_i - y'_j)],$$

and $\binom{n}{2}$ is the total number of pairs and is given by

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

Finally, to identify the best value of α in Equations 1 and 2, we trained several f_θ models with varying α values, i.e., $\alpha \in \{0.01, 0.1, 0.5, 1.0, 5.0\}$, and selected the one that resulted in the highest Kendall's ranking correlation on the validation set.

As in our approach, all MBO methods used for comparison have their surrogate model trained on the offline dataset we have created (see IV-A) for each BCI test case. We evaluated the performance of each model on the held-out points present in the validation set we created, on the respective

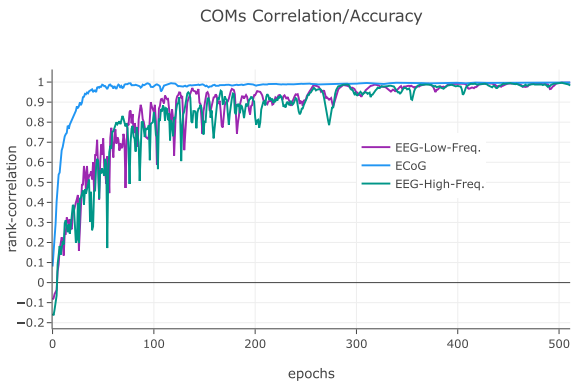

FIGURE 10. Training Curves of COMs [19] surrogate model for BCI test cases. After approximately 512 epochs, the performance index plateaus, at that point the surrogate model has learned to extract meaningful patterns from the offline dataset of each BCI test case.

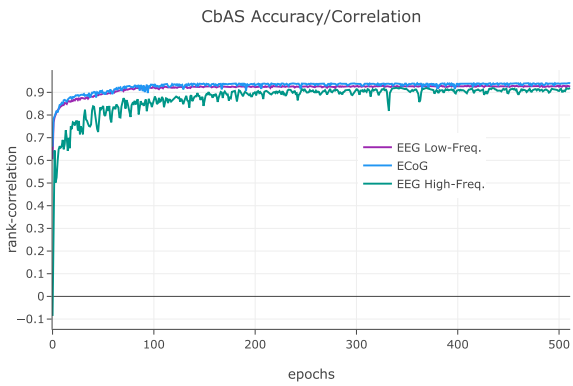

FIGURE 11. Training Curves of CbAS [18] surrogate model for BCI test cases. After approximately 512 epochs, the performance index plateaus, at that point the surrogate model has learned to extract meaningful patterns from the offline dataset of each BCI test case.

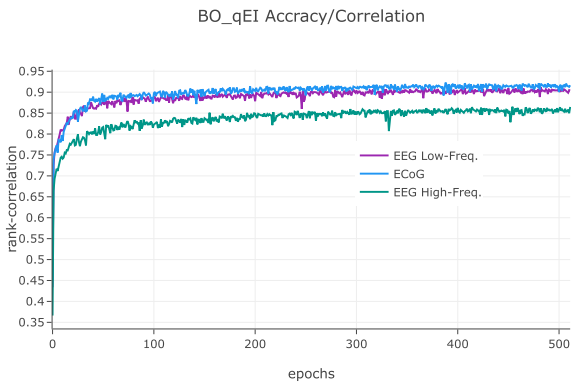

FIGURE 12. Training Curves of BO_qEI [30], [31] surrogate model for BCI test cases. After approximately 512 epochs, the performance index plateaus, at that point the surrogate model has learned to extract meaningful patterns from the offline dataset of each BCI test case.

BCI test case, during each training epoch. Figures 8-12, show the training curve of the surrogate model for the respective MBO method. The performance metric monitored in each plot corresponds to Kendall's rank correlation between the prediction S' of each MBO and the actual ground truth values S for the held-out points on the respective BCI test case at the end of each epoch. Our analysis indicates that we obtained the best model possible, for each method, from the offline data as the training curves converge to an almost perfect rank correlation. Especially our surrogate models (both context-less and context-aware) are the ones that have generalized effectively on the held-out points, acquiring Kendall's ranking correlation coefficient of nearly 1.0. The accuracy obtained by each MBO method, as captured in the respective training curve figure, is aligned with the results presented in Table 11, reflecting the true capability of a method in discovering effective ConvNet architectures for each BCI test case.

V. CONCLUSION AND FUTURE WORK

This paper determines the ideal architectural/training parameter values for the Shallow ConvNet model, the state-of-the-art deep CNN (our baseline model) for decoding the motor imagery EEG/ECoG signals constituting the BCI IV 2a and 4 datasets, respectively. Our findings demonstrate that we can choose better settings for the Shallow ConvNet architectural/training parameters to increase the generalizability and accuracy of the baseline model. More specifically, when we excluded one subject, applied the proposed data-driven optimization method, and then tested the accuracy on that subject, we obtained an improvement of 14.7% and 61.0% on the test-set pertinent to the excluded subject in the BCI IV 2a and 4 datasets, respectively. Finally, when we considered all subjects in our proposed data-driven optimization method, an average improvement of 5.0% and 31.3% in BCI IV 2a and 4 datasets was achieved, respectively.

The proposed optimization methodology keeps the model architecture the same, i.e., the number of layers, the type of layers, and the connectivity between them. Therefore, in the future, we plan to modify the data-driven offline optimization methodology to explore the number, the type, and the connectivity of layers, i.e., a Neural Architecture Search (NAS) algorithm. Moreover, we plan to add hardware implementation constraints in the proposed optimization algorithm, e.g., the Peak Memory Usage (PMU) at inference time, the model size, and the latency to permit the model deployment on low-cost, battery-operated microcontrollers. Finally, except for extracting helpful information directly from the raw EEG/ECoG signals, there are CNN-based analysis techniques that first transform non-stationary signals (such as EEG/ECoG signals) to the frequency domain, and then CNNs extract relevant information from these frequency-domain representations. These kinds of transformations have been applied successfully in induction motor fault diagnosis [32], in the automatic detection of speech deficits in cochlear implant users [33], in phoneme class recognition to extract phone-attribute features [33] and in the classification of noisy non-stationary time-series signals for detecting gravitational-wave signals in intensive real-life [34]. Therefore, one

direction is to first try these approaches in EEG/ECoG motor imagery decoding and optimize the DL models proposed in these works.

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